antineutrons were observed. There now was sufficient evidence for the existence of antiparticles for all three components of the atom. Observations of the antiproton, antineutron and the positron proved the that the notion of charge symmetry was indeed true. If atoms are the building blocks of matter, then antiatoms are the building blocks of antimatter. Now that all the particles that make up an atom were known to have antiparticles, the question remained: was there an antiatom?

Two particle accelerators, the proton synchrotron at CERN and the alternating gradient synchrotron at the Brookhaven National Laboratory both detected the first antinuclei in 1965. Deuterium, a stable isotope of Hydrogen with one neutron and one proton. The atomic nuclei detected was an antideuterium. In 1978, researchers at CERN produced antiprotons from the proton synchrotron, and kept them going round in circles in a machine dubbed the Initial Cooling Experiment, or ICE for a period of 85 hours. It was the first time that anyone, anywhere in the world had stored antiprotons. In 1995, thirty years after the first antinuclei was observed, scientists managed to create the first antiatom in a lab environment. To create an antiatom, it was necessary to have low energy antiparticles, which required specialised equipment. The antiparticles created so far were "hot", with energy levels too high to allow for the creation of antiatoms.

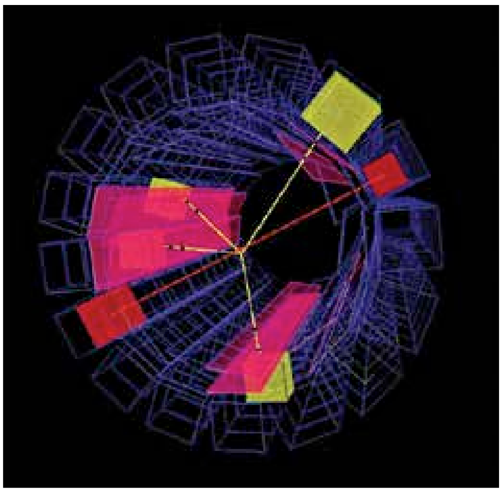
Hydrogen is the simplest atom in the periodic table, and is made up of a proton and an electron. It is the most abundant element in the universe, and comprises nearly 75 percent of the baryonic mass in the universe. Baryonic refers to matter, in other words, the stuff that is not anti-matter. The first antiatom created in a lab environment was the antihydrogen. This was done through an instrument at CERN known as the Low Energy Antiproton Ring. Currently, CERN uses the Antiproton Decelerator to produce antimatter that is of a low enough energy level to allow scientists to study them. In 2003, small quantities of antihelium was produced at Brookhaven National Laboratory. This remains the most complex antiatoms created by humans.

## ANTIMATTER TRAPS

In 2011, CERN managed to store antimatter for a period of sixteen minutes, long enough to start experimenting on the substance, and unlocking its mysteries.

By now, you should have had a clear idea of why antimatter is so rare. It takes an incredible amount of energy to create antimatter, and to store it. Even after the laborious process, only a few antiatoms are created. A feasibility study by NASA to explore the use of antimatter for fuelling spacecraft, pegged the price of creating antimatter at $62.5 trillion a gram, and estimates by CERN puts the figure in the same range. It takes 25 trillion kWh of electricity to make a single gram of antimatter. CERN has produced less than 10 nanograms of antimatter so far, and has the capability to create about one billionth of a gram every year. CERN would take 1 billion years to produce one gram of antimatter. Antimatter cannot be stored in a normal container, as matter and antimatter mutually annihilate each other on contact, usually producing gamma rays and neutrinos in the process. Researchers do not even know that they have created antimatter till they observe the annihilation event.

Credit: CERN



An antihydrogen annihilation event in the ATHENA experiment. The pink silicon microchips indicate four pairs of quarks and antiquarks, the yellow cubes show captured energy and the red line denotes the gamma rays produced by the annihilation.

Antiparticles are stored by suspension in vacuum through electric fields, or if the antimatter has no charge, through superconducting magnets. Researchers are developing magnetic bottles and optical traps using lasers to store antimatter.

## ANTIMATTER IN NATURE

Conditions similar to that of particle accelerators are created in nature as well, including pulsars and the magnetic fields of celestial bodies. A supermassive black hole accelerates particles in a jet stream that interacts with the gas clouds of two colliding galaxies, Abell 3411 and Abell 3412 to create a natural particle accelerator. Naturally occurring antiparticles have been detected in the Van Allen Belts around Earth. If there are entire galaxies made up of antimatter, they would be indistinguishable from regular galaxies through our telescopes. If such galaxies exist, they would be from the early days of the big bang, near our horizon of visibility.

The radiation from unstable atomic nuclei also produce small quantities of antimatter in some cases. Human beings produce antimatter by eating, drinking and breathing. This is because of the potassium-40 in the body, and a person weighing 80 kg produces about 180 positrons every hour.

The big bang actually created equal amounts of matter and antimatter. Soon after though, most of the antimatter disappeared, leaving behind the matter. Theoretically, there should be the same amount of matter and antimatter in the universe. Why humans can observe far more matter than antimatter remains a mystery.